Python for Data — 14-Week Study Plan

Based on Jess Ramos' 3-Level Roadmap · 8 Projects · ~10–12 hrs/week

14	8	3	6-fig
total weeks	real projects	skill levels	salary goal

Level 1 — The Basics (Weeks 1–3) Building confidence with data exploration

Wk 1	Python & pandas foundations	Setup + data types + reading files
Mon	☐ Install Python, VS Code / Jupyter, create your virtual environment	
Tue	☐ Learn Python basics: variables, lists, dicts, loops, functions	
Wed	☐ Import pandas & numpy; read a CSV with pd.read_csv(); inspect shape, dtypes, head	
Thu	☐ Change data types (.astype()), rename columns, basic .info() & .describe()	
Fri	☐ Read an Excel file with pd.read_excel(); practice on 2 different datasets	
Weekend	☐ Mini-challenge: load the Superstore CSV, fix column names, inspect all data types ***Project 1 — start***	

Wk 2	Selection, filtering & string ops	loc/iloc · conditions · string methods
Mon	☐ Select columns with df['col'] and df[['a','b']]; select rows with .loc[] and .iloc[]	
Tue	☐ Filter rows with boolean conditions (&, |, ~), .isin(), and .between()	
Wed	☐ String methods: .str.lower(), .str.strip(), .str.contains(), .str.replace()	
Thu	☐ Handle missing values: .isnull(), .fillna(), .dropna()	
Fri	☐ Practice: filter the Superstore data by region and product category, clean string columns	
Weekend	☐ Load the Student Grades Excel file; fill/drop nulls; filter by score range ***Project 2 — start***	

Wk 3	Basic charts & project delivery	matplotlib · bar · line · finish projects 1 & 2
Mon	☐ matplotlib basics: bar chart, line chart, titles, axis labels, colors	
Tue	☐ Plot top 10 products by revenue for Superstore — add annotations	
Wed	☐ Complete Project 1: Superstore Sales Explorer — full notebook, clean code, comments ***Project 1 — DONE***	
Thu	☐ Bar-chart grade distribution for Student Grades dataset	
Fri	☐ Complete Project 2: Student Grades Cleaner — polish notebook, push to GitHub ***Project 2 — DONE***	

Weekend	☐ Level 1 review: redo one task from each week from scratch without notes

> *Milestone: 2 projects on GitHub. You now have a real portfolio starting point.*

Level 2 — Intermediate (Weeks 4–8) Solving real business problems

Wk 4	**groupby & aggregation**	groupby · agg · pivot tables
Mon	☐ .groupby() with .sum(), .mean(), .count(), .max()	
Tue	☐ Multi-column groupby; .agg({'col': ['sum','mean']}) named aggregations	
Wed	☐ Pivot tables with pd.pivot_table(); reset index; sort results	
Thu	☐ Detect and handle duplicates: .duplicated(), .drop_duplicates()	
Fri–Wknd	☐ Start Project 3: group orders by month and customer segment, compute totals ***Project 3 — start***	

Wk 5	**Merging & joining datasets**	merge · join · concat
Mon	☐ pd.merge(): inner, left, right, outer joins — understand the difference	
Tue	☐ Merge on multiple keys; handle suffixes for overlapping column names	
Wed	☐ pd.concat() for stacking rows; validate merge results with shape checks	
Thu	☐ Join orders + customers tables for Project 3; compute MoM revenue growth %	
Fri–Wknd	☐ Complete Project 3: Monthly Revenue Dashboard with multi-panel chart ***Project 3 — DONE***	

Wk 6	**Date/time & conditional logic**	datetime · np.where · pd.cut
Mon	☐ Parse dates with pd.to_datetime(); extract year, month, day, weekday	
Tue	☐ Compute timedeltas: days between dates, days-since-last-purchase	
Wed	☐ Conditional logic: np.where(), np.select(), pd.cut() for binning	
Thu	☐ Create new derived columns: tenure buckets, churn-risk flags, revenue tiers	
Fri–Wknd	☐ Start Project 4: parse order dates, calculate days-since-purchase per customer ***Project 4 — start***	

Wk 7	**Sorting, ranking & stats charts**	rank · sort · boxplot · histogram
Mon	☐ .sort_values(), .rank() with different methods (dense, min, percent)	
Tue	☐ Histograms and boxplots with matplotlib/seaborn; interpret distributions	
Wed	☐ Scatter plots and correlation heatmaps with seaborn	

Thu	☐ Complete Project 4: rank customers by recency, visualize churn-risk distribution ***Project 4 — DONE***
Fri–Wknd	☐ Start Project 5: merge employee + department tables, remove duplicates ***Project 5 — start***

Wk 8	Project 5 & Level 2 review	HR analysis · consolidate Level 2 skills
Mon–Tue	☐ Segment employees by tenure buckets; compute attrition rate per department	
Wed	☐ Build statistical summary report; produce grouped bar chart by department	
Thu	☐ Complete and polish Project 5: HR Attrition Analysis ***Project 5 — DONE***	
Fri–Wknd	☐ Level 2 review: rebuild one aggregation + one merge task from scratch; update GitHub READMEs	

Milestone: 5 projects on GitHub. You can now solve real business analyst interview tasks.

Level 3 — Scalable Analysis (Weeks 9–14) The 'big bucks' level

Wk 9	Large datasets & reusable functions	chunked reads · functions · modules
Mon	☐ Read large CSVs in chunks with chunksize; use dtype param to reduce memory	
Tue	☐ Write reusable functions: cleaning function, aggregation function, formatter	
Wed	☐ Organize code into a .py module; import and call from a notebook	
Thu	☐ Write docstrings; add type hints; unit test one function with assert	
Fri–Wknd	☐ Start Project 6: scaffold the automated report script — ingest + clean + compute KPIs ***Project 6 — start***	

Wk 10	Data quality & QA checks	validation · assertions · logging
Mon	☐ Write QA functions: null-rate check, range check, uniqueness check	
Tue	☐ Add logging with Python's logging module to track pipeline steps	
Wed	☐ Build a QA summary report that flags issues before the main analysis runs	
Thu	☐ Integrate QA suite into Project 6 report script	
Fri–Wknd	☐ Add HTML output generation to Project 6; test with fresh CSV file ***Project 6 — cont.***	

Wk 11	Automation & scheduling	scheduling · email · automation
Mon	☐ Automate file ingestion: watch a folder for new CSV files with pathlib	
Tue	☐ Send automated emails with smtplib / yagmail including HTML report attachment	

Wed	☐ Schedule script with Windows Task Scheduler or cron (Linux/Mac)	
Thu–Fri	☐ Complete Project 6: run end-to-end, verify automated email works	***Project 6 — DONE***
Weekend	☐ Start Project 7: load time-series sales data; compute rolling averages	***Project 7 — start***

Wk 12	Trend analysis & advanced charts	Plotly · forecasting · confidence bands
Mon	☐ Install Plotly; build interactive line and bar charts with hover tooltips	
Tue	☐ Rolling averages, exponential smoothing; detect seasonality visually	
Wed	☐ Simple forecasting: linear trend line with numpy.polyfit + confidence interval shading	
Thu–Fri	☐ Build reusable plot_trend(df, col) function; export as shareable HTML notebook	
Weekend	☐ Complete Project 7: Sales Trend Forecasting Tool	***Project 7 — DONE***

Wk 13	End-to-end pipeline — build	multi-source ingestion · full transform
Mon	☐ Define pipeline architecture: ingest → QA → transform → output (draw it out first)	
Tue	☐ Ingest 3+ source files; run QA suite; merge into one master dataset	
Wed	☐ Run all complex transformations using reusable functions from Week 9	
Thu	☐ Build Plotly subplots dashboard: KPI cards + trend chart + segment breakdown	
Fri–Wknd	☐ Write executive summary section; export full dashboard to HTML	***Project 8 — cont.***

Wk 14	Polish, publish & career prep	GitHub · README · portfolio storytelling
Mon	☐ Complete and stress-test Project 8 with a fresh dataset	***Project 8 — DONE***
Tue	☐ Write a strong README for each project: problem, approach, key findings, screenshots	
Wed	☐ Create a portfolio landing page (GitHub profile README) linking all 8 projects	
Thu	☐ Practice explaining 3 projects out loud as if in an interview — time yourself	
Fri	☐ Update LinkedIn with projects; apply to 3 data analyst roles	
Weekend	☐ Celebrate — you have a complete, job-ready Python data portfolio!	

> *Final milestone: 8 projects, a GitHub portfolio, and real-world analytical thinking at all 3 levels. Time to apply!*

Tips for success

- Daily commitment: aim for 1–2 hrs on weekdays, 3–4 hrs on weekends for project work (~10–12 hrs/week)
- Use real Kaggle datasets for every project — not toy examples. Kaggle's 'Sample Superstore' is great for projects 1, 3.

- For Level 3, try to use data from your own field or job if possible — it makes your portfolio far more compelling.
- After each project, write a short paragraph explaining what you did and what you found, as if presenting to a manager.
- Build a GitHub portfolio as you go — it's proof of your progression to future employers.